

October 31, 2014

MDEQ, WRD-PSM, Grand Rapids  
Attn: Ms. Chris Little  
350 Ottawa, NW, Unit 10  
Grand Rapids, MI 48909

**Project: Wastewater Surveys 2014**

Dear Ms. Chris Little,

Enclosed is a copy of the laboratory report for the following work order(s) received by TriMatrix Laboratories:

| <b>Work Order</b> | <b>Received</b> | <b>Description</b>      |
|-------------------|-----------------|-------------------------|
| 1410434           | 10/21/2014      | MI0020257/Cadillac WWTP |

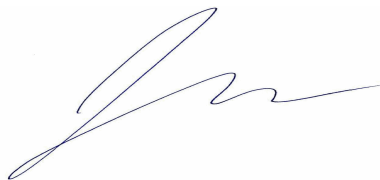
This report relates only to the sample(s) as received. Test results are in compliance with the requirements of the National Environmental Laboratory Accreditation Program (NELAP) and/or one of the following certification programs:

ACCLASS DoD-ELAP/ISO17025 (#ADE-1542); Arkansas DEP (#88-0730/13-049-0); Florida DEP (#E87622-24); Georgia EPD (#E87622-24); Illinois DEP (#200026/003329); Kansas DPH (#E-10302); Kentucky DEP (#0021); Louisiana DEP (#103068); Michigan DPH (#0034); Minnesota DPH (#491715); New York ELAP (#11776/48855); North Carolina DNRE (#659); Texas CEQ (#T104704495-14-4); Virginia DCLS (#460153/2592); Wisconsin DNR (#999472650); USDA Soil Import Permit (#P330-12-00236).

Any qualification or narration of results, including sample acceptance requirements and test exceptions to the above referenced programs, is presented in the Statement of Data Qualifications and Project Technical Narrative sections of this report. Estimates of analytical uncertainties and certification documents for the test results contained within this report are available upon request.

If you have any questions or require further information, please do not hesitate to contact me.

Sincerely,



James D. McFadden  
Project Chemist

**PROJECT TECHNICAL NARRATIVE(s)**

No Project Narrative is associated with this report.

**STATEMENT OF DATA QUALIFICATIONS****Physical/Chemical Parameters by EPA/APHA/ASTM Methods**

**Qualification:** Sample was collected in a clear plastic 50mL vial. Insufficient sample for matrix QC.

Analysis: OIA-1677-09

Sample/Analyte: 1410434-01

CD-001A-1

Cyanide, Available

## ANALYTICAL REPORT

Client: **MDEQ, WRD-PSM, Grand Rapids**  
 Project: Wastewater Surveys 2014  
 Client Sample ID: **CD-001A-1**  
 Lab Sample ID: **1410434-01**  
 Matrix: Waste Water

Work Order: **1410434**  
 Description: MI0020257/Cadillac WWTP  
 Sampled: 10/14/14 10:15  
 Sampled By: C.Little  
 Received: 10/21/14 18:00

### Physical/Chemical Parameters by EPA/APHA/ASTM Methods

| Analyte             | Analytical Result | RL     | MDL     | Unit | Dilution Factor | Method      | Date Time Analyzed | By  | QC Batch |
|---------------------|-------------------|--------|---------|------|-----------------|-------------|--------------------|-----|----------|
| *Cyanide, Available | ND U              | 0.0020 | 0.00033 | mg/L | 1               | OIA-1677-09 | 10/28/14 10:20     | LMA | 1411974  |

\*See Statement of Data Qualifications

## QUALITY CONTROL REPORT

### Physical/Chemical Parameters by EPA/APHA/ASTM Methods

| QC Type | Sample Conc. | Spike Qty. | Result | Unit | Spike % Rec. | Control Limits | RPD | RPD Limits | RL | MDL |
|---------|--------------|------------|--------|------|--------------|----------------|-----|------------|----|-----|
|---------|--------------|------------|--------|------|--------------|----------------|-----|------------|----|-----|

**Analyte: Cyanide, Available/OIA-1677-09**

QC Batch: 1411974 (Method Specific Preparation)

Analyzed: 10/28/2014 By: LMA

|                           |  |        |               |      |     |        |  |  |        |  |
|---------------------------|--|--------|---------------|------|-----|--------|--|--|--------|--|
| Method Blank              |  |        | ND U          | mg/L |     |        |  |  | 0.0020 |  |
| Laboratory Control Sample |  | 0.0200 | <b>0.0201</b> | mg/L | 101 | 82-132 |  |  | 0.0020 |  |

## MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

## ENVIRONMENTAL LABORATORY - ANALYSIS REQUEST SHEET

1410434

Lab Work Order Number

Project Name

Analyze

1410104

Cadillac WWTP

WASTE WATER

Site Code/Project Number

MI0020257

AY

15

CC Email 1 to send additional reports to

Project TAT Days

Sample Collector

Chris C. Little

Dept/Division/District

MDEQ-WRD-PSM-GR

Index

63206

CC Email 2 to send additional reports to

Project Due Date

Sample Collector Phone

cell 616-550-7668

State Project Manager

Chris C. Little

PCA

42203

CC Email 3 to send additional reports to

Accept Analysis hold time codes Y/N/No

Contract Firm

State Project Manager Email

littlec@michigan.gov

Project

481000

Overlook Lab Choice 1

Tri Matrix

Contract Firm Primary Contact

State Project Manager Phone

cell 616-550-7668

Phone

00

Overlook Lab Choice 2

Primary Contact Phone

Lab Use Only

Field Sample Identification

Collection Date

Collection Time

Container Count

Comments

1

01

CD-001A

10/15/14

1010

2

02

CD-001A-1

10/14/14

1015

3

03

CD-UP

10/14/14

1100

4

04

CD-001A-F

10/15/14

0700

5

6

7

8

9

10

## MA - TOTAL METALS

Cooper - Cu  
Hardness - Ca, Mg2 3 4 5 6 7 8 9 10  
2 3 4 5 6 7 8 9 10

Shared bottle for line 1

## GENERAL CHEMISTRY

GCN - General Cyanide Available Cyanide - CN

1 2 3 4 5 6 7 8 9 10

GHN - General Chemistry Neutral

Suspended Solids - SS

1 2 3 4 5 6 7 8 9 10

MNI - Inorganic Nitrate Neutral

Total Alkalinity

1 2 3 4 5 6 7 8 9 10

GA - General Chemistry Anions

Ammonia - NH<sub>3</sub>

1 2 3 4 5 6 7 8 9 10

Total Phosphorus - TP

1 2 3 4 5 6 7 8 9 10

| Chain of Custody  | Relinquished by   |                                   | Received By       |                      | Date / Time   |
|-------------------|-------------------|-----------------------------------|-------------------|----------------------|---------------|
|                   | Print Name & Org. | Chris C. Little - MDEQ/WRD/PSM/GR | Print Name & Org. | Melissa Smith        | 10/15/14 1431 |
|                   | Signature:        | <i>Chris C. Little</i>            | Signature:        | <i>Melissa Smith</i> |               |
|                   | Print Name & Org. |                                   | Print Name & Org. | 2100 TRINITY         | 10/21/14 1800 |
| Signature:        |                   | Signature:                        |                   |                      |               |
| Print Name & Org. |                   | Print Name & Org.                 |                   |                      |               |
| Signature:        |                   | Signature:                        |                   |                      |               |



# SAMPLE RECEIVING / LOG-IN CHECKLIST



|                                          |                                        |
|------------------------------------------|----------------------------------------|
| Client: <u>NDEP</u>                      | Work Order #: <u>141010 1410434</u>    |
| Receipt Record Page/Line #: <u>12-29</u> | Project Chemist: _____ Sample #: _____ |

|                                                 |                                                                                                                    |                        |                                                                                                                                                     |                         |                                                                 |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------|
| Recorded by (initials/date): <u>JN 10/21/14</u> | <input checked="" type="checkbox"/> Cooler<br><input type="checkbox"/> Box<br><input type="checkbox"/> Other _____ | Qty Received: <u>1</u> | <input checked="" type="checkbox"/> IR Gun (#202)<br><input type="checkbox"/> Digital Thermometer (#54)<br><input type="checkbox"/> Other (# _____) | Thermometer Used: _____ | <input type="checkbox"/> See Additional Cooler Information Form |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------|

| Cooler #                                                                                                                                                                    | Time                 | Cooler #                                                                                                                                                         | Time       | Cooler #                                                                                                                                                         | Time                 | Cooler #                                                                                                                                                         | Time |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <u>113478</u>                                                                                                                                                               | <u>2237</u>          |                                                                                                                                                                  |            |                                                                                                                                                                  |                      |                                                                                                                                                                  |      |
| Custody Seals:<br><input checked="" type="checkbox"/> None<br><input type="checkbox"/> Present / Intact<br><input type="checkbox"/> Present / Not Intact                    |                      | Custody Seals:<br><input type="checkbox"/> None<br><input type="checkbox"/> Present / Intact<br><input type="checkbox"/> Present / Not Intact                    |            | Custody Seals:<br><input type="checkbox"/> None<br><input type="checkbox"/> Present / Intact<br><input type="checkbox"/> Present / Not Intact                    |                      | Custody Seals:<br><input type="checkbox"/> None<br><input type="checkbox"/> Present / Intact<br><input type="checkbox"/> Present / Not Intact                    |      |
| Coolant Type:<br><input checked="" type="checkbox"/> Loose Ice<br><input type="checkbox"/> Bagged Ice<br><input type="checkbox"/> Blue Ice<br><input type="checkbox"/> None |                      | Coolant Type:<br><input type="checkbox"/> Loose Ice<br><input type="checkbox"/> Bagged Ice<br><input type="checkbox"/> Blue Ice<br><input type="checkbox"/> None |            | Coolant Type:<br><input type="checkbox"/> Loose Ice<br><input type="checkbox"/> Bagged Ice<br><input type="checkbox"/> Blue Ice<br><input type="checkbox"/> None |                      | Coolant Type:<br><input type="checkbox"/> Loose Ice<br><input type="checkbox"/> Bagged Ice<br><input type="checkbox"/> Blue Ice<br><input type="checkbox"/> None |      |
| Coolant Location: _____                                                                                                                                                     |                      | Coolant Location: _____                                                                                                                                          |            | Coolant Location: _____                                                                                                                                          |                      | Coolant Location: _____                                                                                                                                          |      |
| Dispersed / Top / Middle / Bottom                                                                                                                                           |                      | Dispersed / Top / Middle / Bottom                                                                                                                                |            | Dispersed / Top / Middle / Bottom                                                                                                                                |                      | Dispersed / Top / Middle / Bottom                                                                                                                                |      |
| Temp Blank Present: <input type="checkbox"/> Yes <input type="checkbox"/> No                                                                                                |                      | Temp Blank Present: <input type="checkbox"/> Yes <input type="checkbox"/> No                                                                                     |            | Temp Blank Present: <input type="checkbox"/> Yes <input type="checkbox"/> No                                                                                     |                      | Temp Blank Present: <input type="checkbox"/> Yes <input type="checkbox"/> No                                                                                     |      |
| If Present, Temperature Blank Location is: _____                                                                                                                            |                      | If Present, Temperature Blank Location is: _____                                                                                                                 |            | If Present, Temperature Blank Location is: _____                                                                                                                 |                      | If Present, Temperature Blank Location is: _____                                                                                                                 |      |
| <input type="checkbox"/> Representative <input type="checkbox"/> Not Representative                                                                                         |                      | <input type="checkbox"/> Representative <input type="checkbox"/> Not Representative                                                                              |            | <input type="checkbox"/> Representative <input type="checkbox"/> Not Representative                                                                              |                      | <input type="checkbox"/> Representative <input type="checkbox"/> Not Representative                                                                              |      |
| Observed °C                                                                                                                                                                 | Correction Factor °C | Actual °C                                                                                                                                                        |            | Observed °C                                                                                                                                                      | Correction Factor °C | Actual °C                                                                                                                                                        |      |
| Temp Blank:                                                                                                                                                                 |                      |                                                                                                                                                                  |            | Temp Blank:                                                                                                                                                      |                      |                                                                                                                                                                  |      |
| Sample 1:                                                                                                                                                                   | <u>1.6</u>           | <u>0</u>                                                                                                                                                         | <u>1.6</u> | Sample 1:                                                                                                                                                        |                      |                                                                                                                                                                  |      |
| Sample 2:                                                                                                                                                                   | <u>1.3</u>           | <u>0</u>                                                                                                                                                         | <u>1.3</u> | Sample 2:                                                                                                                                                        |                      |                                                                                                                                                                  |      |
| Sample 3:                                                                                                                                                                   | <u>1.4</u>           | <u>0</u>                                                                                                                                                         | <u>1.4</u> | Sample 3:                                                                                                                                                        |                      |                                                                                                                                                                  |      |
| 3 Sample Average °C: <u>1.4</u>                                                                                                                                             |                      |                                                                                                                                                                  |            | 3 Sample Average °C: _____                                                                                                                                       |                      |                                                                                                                                                                  |      |
| <input type="checkbox"/> Cooler ID on COC?<br><input type="checkbox"/> VOC Trip Blank received?                                                                             |                      |                                                                                                                                                                  |            | <input type="checkbox"/> Cooler ID on COC?<br><input type="checkbox"/> VOC Trip Blank received?                                                                  |                      |                                                                                                                                                                  |      |

If any shaded areas checked, complete Sample Receiving Non-Conformance and/or Inventory Form

## Paperwork Received

|                                                                        |                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> Yes<br><input type="checkbox"/> No | <input type="checkbox"/> Chain of Custody record(s)? If No, Initiated By: _____<br><input type="checkbox"/> Received for Lab Signed/Date/Time: _____<br><input type="checkbox"/> Shipping document? <u>FOR SHEET</u><br><input type="checkbox"/> Other: _____ |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## COC Information

|                                                                                                  |
|--------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> TriMatrix COC<br><input checked="" type="checkbox"/> Other: <u>NDEP</u> |
| COC ID Numbers: <u>13366</u><br><u>13367</u><br><u>13368</u>                                     |

## Check COC for Accuracy

|                                                                        |                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> No | <input type="checkbox"/> Analysis Requested?<br><input checked="" type="checkbox"/> Sample ID matches COC?<br><input type="checkbox"/> Sample Date and Time matches COC?<br><input type="checkbox"/> Container type completed on COC?<br><input checked="" type="checkbox"/> All container types indicated are received? |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Sample Condition Summary

|                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> N/A<br><input checked="" type="checkbox"/> Yes<br><input type="checkbox"/> No | <input type="checkbox"/> Broken containers/lids?<br><input type="checkbox"/> Missing or incomplete labels?<br><input type="checkbox"/> Illegible information on labels?<br><input type="checkbox"/> Low volume received?<br><input type="checkbox"/> Inappropriate or non-TriMatrix containers received?<br><input type="checkbox"/> VOC vials / TOX containers have headspace?<br><input type="checkbox"/> Extra sample locations / containers not listed on COC? |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Check Sample Preservation

|                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> N/A<br><input checked="" type="checkbox"/> Yes<br><input type="checkbox"/> No | <input type="checkbox"/> Temperature Blank OR average sample temperature, ≥26° C?<br><input type="checkbox"/> If either is ≥26° C, was thermal preservation required?<br>If "Yes", Project Chemist Approval Initials: _____<br>If "Yes" Completed Non Con Cooler - Cont Inventory Form?<br><input checked="" type="checkbox"/> Completed Sample Preservation Verification Form?<br><input checked="" type="checkbox"/> Samples chemically preserved correctly?<br>If "No", added orange tag?<br><input type="checkbox"/> Received pre-preserved VOC soils?<br><input type="checkbox"/> MeOH <input type="checkbox"/> Na <sub>2</sub> SO <sub>4</sub> |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Check for Short Hold-Time Prep/Analyses

|                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Bacteriological<br><input type="checkbox"/> Air Bags<br><input type="checkbox"/> EnCores / Methanol Pre-Preserved<br><input type="checkbox"/> Formaldehyde/Aldehyde<br><input type="checkbox"/> Green-tagged containers<br><input type="checkbox"/> Yellow/White-tagged 1 L ambers (SV Prep-Lab) | <b>AFTER HOURS ONLY:</b><br>COPIES OF COC TO LAB AREA(S)<br><input checked="" type="checkbox"/> NONE RECEIVED<br><input type="checkbox"/> RECEIVED, COCs TO LAB(S) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Notes

|                                                                                                    |                                                  |                                   |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------|
| <input type="checkbox"/> Trip Blank received <input type="checkbox"/> Trip Blank not listed on COC |                                                  |                                   |
| Cooler Received (Date/Time): <u>JN 10/21/14</u>                                                    | Paperwork Delivered (Date/Time): <u>10/21/14</u> | ≤1 Hour Goal Met? <u>Yes / No</u> |